

Chapter 56

Conservation Biology and Restoration Ecology

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: Striking Gold

- 1.8 million species have been named and described.
- Biologists estimate 10–200 million species exist on Earth.
- Tropical forests contain some of the greatest concentrations of species and are being destroyed at an alarming rate.
- Humans are rapidly pushing many species toward extinction.

Tropical deforestation in West Kalimantan, Indonesian



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- **Conservation biology** seeks to preserve life and integrates several fields:
 - *Ecology*
 - *Physiology*
 - *Molecular biology*
 - *Genetics*
 - *Evolutionary biology*
 - **Restoration ecology** applies ecological principles to return degraded ecosystems to conditions as similar as possible to their natural state.

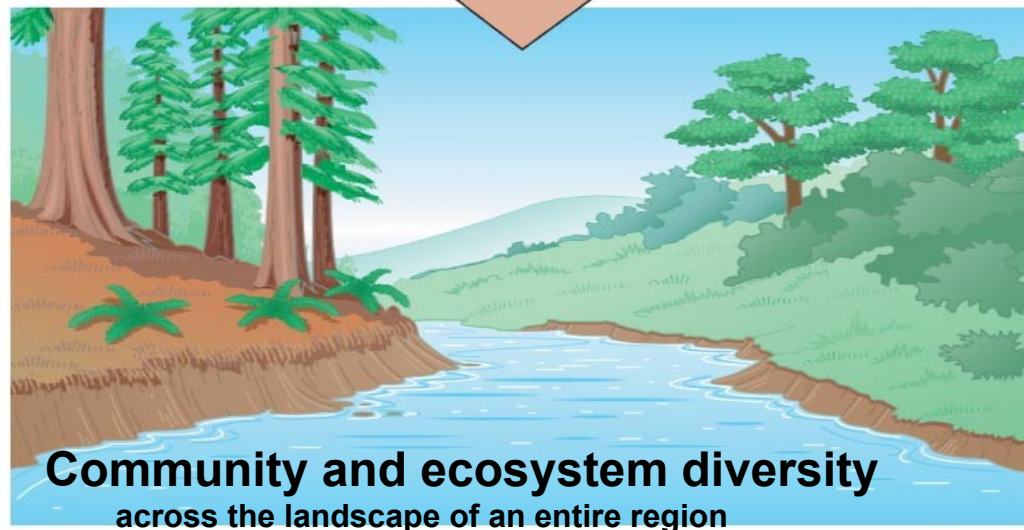
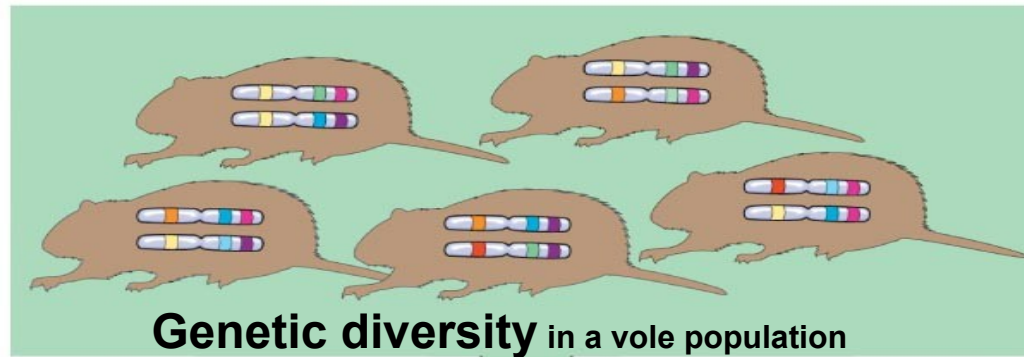
Human activities threaten Earth's biodiversity

- Rates of species extinction are difficult to determine under natural conditions.
- The high rate of species extinction is largely a result of ecosystem degradation by humans.
- Humans are threatening Earth's biodiversity.

Three Levels of *Biodiversity*

- Biodiversity has three main components:
 - *Genetic diversity*
 - *Species diversity*
 - *Ecosystem diversity*

Three Levels of Biodiversity



Genetic Diversity & Species Diversity

- **Genetic diversity** comprises genetic variation within a population and between populations.
- **Species diversity** is the variety of species in an ecosystem or throughout the biosphere.
- According to the U.S. Endangered Species Act:
 - An **endangered species** is “in danger of becoming extinct throughout all or a significant portion of its range”
 - A **threatened species** is likely to become endangered in the foreseeable future.

Ecosystem Diversity

- Human activity is reducing ecosystem diversity, the variety of ecosystems in the biosphere.
- More than 50% of wetlands in the contiguous United States have been drained and converted to other ecosystems.

The endangered Marianas “flying fox” bat is an important pollinator



Biodiversity and Human Welfare

- Human **biophilia** allows us to recognize the **value of biodiversity for its own sake**.
- **Species diversity** brings humans **practical benefits**. The enormous genetic diversity of organisms has potential for great human benefit.
- In the United States, 25% of **prescriptions** contain substances originally derived from plants.
- For example, the rosy periwinkle contains alkaloids that inhibit cancer growth.

The rosy periwinkle, a plant that saves lives



Ecosystem Services

- **Ecosystem services** encompass all the processes through which natural ecosystems and their species **help sustain human life**.
- Some examples of ecosystem services:
 - ***Purification of air and water***
 - ***Detoxification and decomposition of wastes***
 - ***Cycling of nutrients***
 - ***Moderation of weather extremes***.

Three Threats to Biodiversity

- Most species loss can be traced to three major threats:
 - **Habitat destruction**
 - **Introduced species**
 - **Overexploitation**
- Human alteration of habitat is the greatest threat to biodiversity throughout the biosphere.
- In almost all cases, habitat fragmentation and destruction lead to loss of biodiversity.

Introduced Species

- **Introduced species** are those that humans move from native locations to new geographic regions.
- Without their native predators, parasites, and pathogens, introduced species may **spread rapidly**.
- Introduced species that gain a foothold in a new habitat usually **disrupt** their adopted **community**.
- Sometimes humans introduce species by accident, as in case of the brown tree snake arriving in Guam as a cargo ship “stowaway.”

Overexploitation

- **Overexploitation** is *human harvesting* of wild plants or animals *at rates exceeding the ability of populations* of those species *to rebound*.
- Overexploitation by the fishing industry has greatly reduced populations of some game fish, such as bluefin tuna.
- DNA analysis can help conservation biologists to identify the source of illegally obtained animal products.

Overexploitation



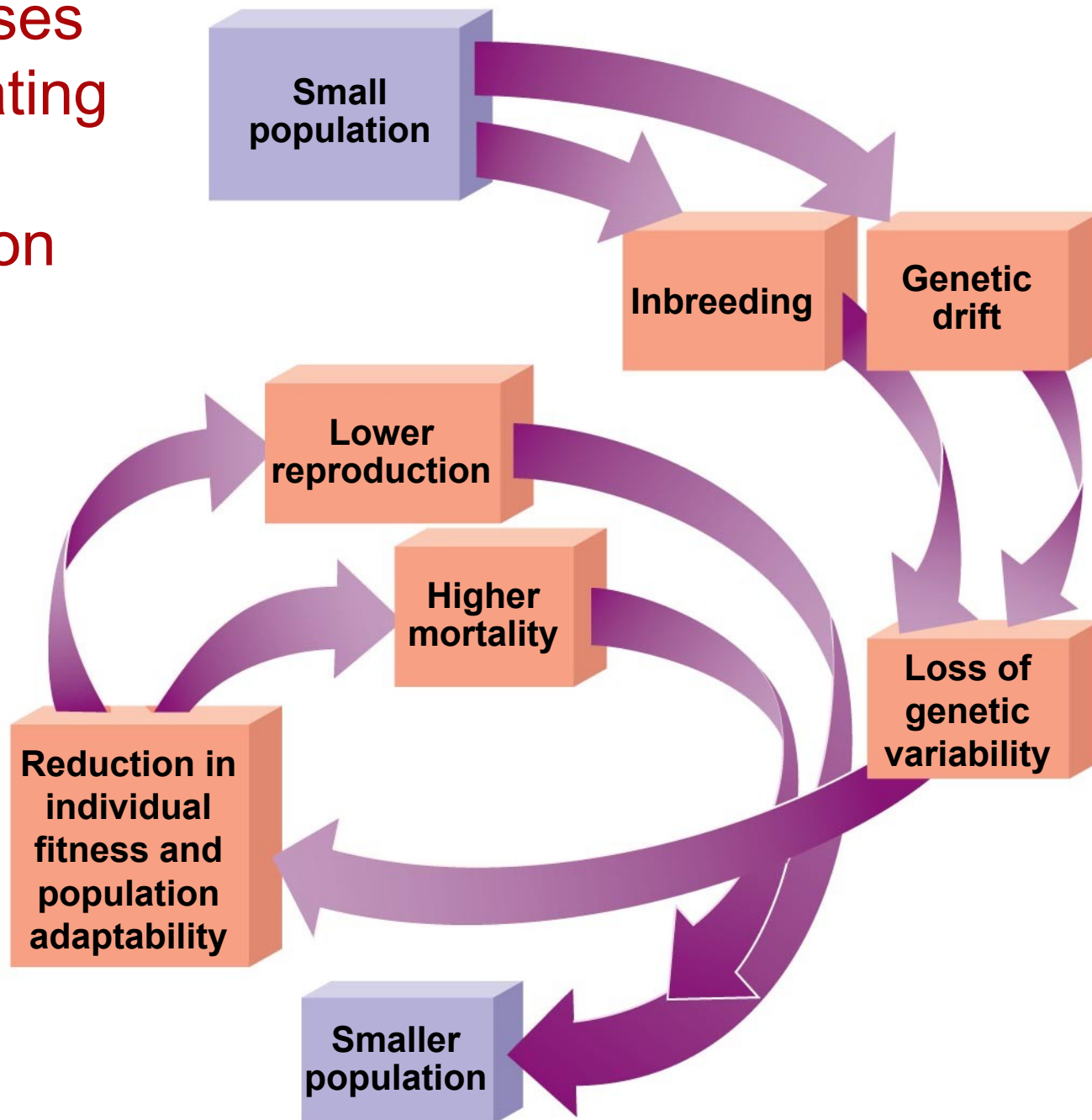
Population conservation focuses on population size, genetic diversity, and critical habitat

- Biologists focusing on conservation at the population and species levels follow two main approaches:
 - The small-population approach
 - The declining-population approach

Small-Population Approach

- The small-population approach studies processes that can make small populations become extinct.
- A small population is prone to positive-feedback loops that draw it down an **extinction vortex**.
- The key factor driving the extinction vortex is **loss of the genetic variation necessary to enable evolutionary responses to environmental change**.

Processes culminating in an extinction vortex



Minimum Viable Population Size

- **Minimum viable population MVP** is the minimum population size at which a **species can survive**.
- The MVP depends on factors that affect a population's chances for survival over a particular time.
- A meaningful estimate of MVP requires determining the **effective population size**, which is based on the population's **breeding potential**.

Case Study: *Analysis of Grizzly Bear Populations*

- One of the first population viability analyses was conducted as part of a long-term study of grizzly bears in Yellowstone National Park.
- This grizzly population is about 400, but the N_e is about 100.
- The Yellowstone grizzly population has low genetic variability compared with other grizzly populations.
- Introducing individuals from other populations would increase the numbers and genetic variation.

Long-term monitoring of a grizzly bear population



Declining-Population Approach

- The declining-population approach
 - Focuses on threatened and endangered populations that show a downward trend, regardless of population size.
 - Emphasizes the environmental factors that caused a population to decline.

Steps for Analysis and Intervention

- The declining-population approach involves several steps:
 - Confirm that the population is in decline
 - Study the species' natural history
 - Develop hypotheses for all possible causes of decline
 - Test the hypotheses in order of likeliness
 - Apply the results of the diagnosis to manage for recovery.

Weighing Conflicting Demands

- Conserving species often requires resolving **conflicts between habitat needs** of endangered species and **human demands**.
- For example, in the U.S. Pacific Northwest, habitat preservation for many species is at odds with timber and mining industries.
- Managing habitat for one species might have positive or negative effects on other species.

Landscape and regional conservation aim to sustain entire biotas

- Conservation biology has attempted to sustain the biodiversity of entire communities, ecosystems, and landscapes.
- **Ecosystem management** is part of landscape ecology, which seeks to **make biodiversity conservation part of land-use planning**.
- The structure of a landscape can strongly influence biodiversity.

Fragmentation and Edges

- The boundaries, or edges, between ecosystems are defining features of landscapes.
- Some species take advantage of edge communities to access resources from both adjacent areas.
- The Biological Dynamics of Forest Fragments Project in the Amazon examines the effects of fragmentation on biodiversity.
- Landscapes dominated by fragmented habitats support fewer species due to a loss of species adapted to habitat interiors.

Edges between ecosystems



(a) Natural edges



(b) Edges created by human activity

Corridors That Connect Habitat Fragments

- A **movement corridor** is a narrow strip of quality habitat connecting otherwise isolated patches.
- Movement corridors promote dispersal and help sustain populations.
- In areas of heavy human use, artificial corridors are sometimes constructed.

An artificial corridor



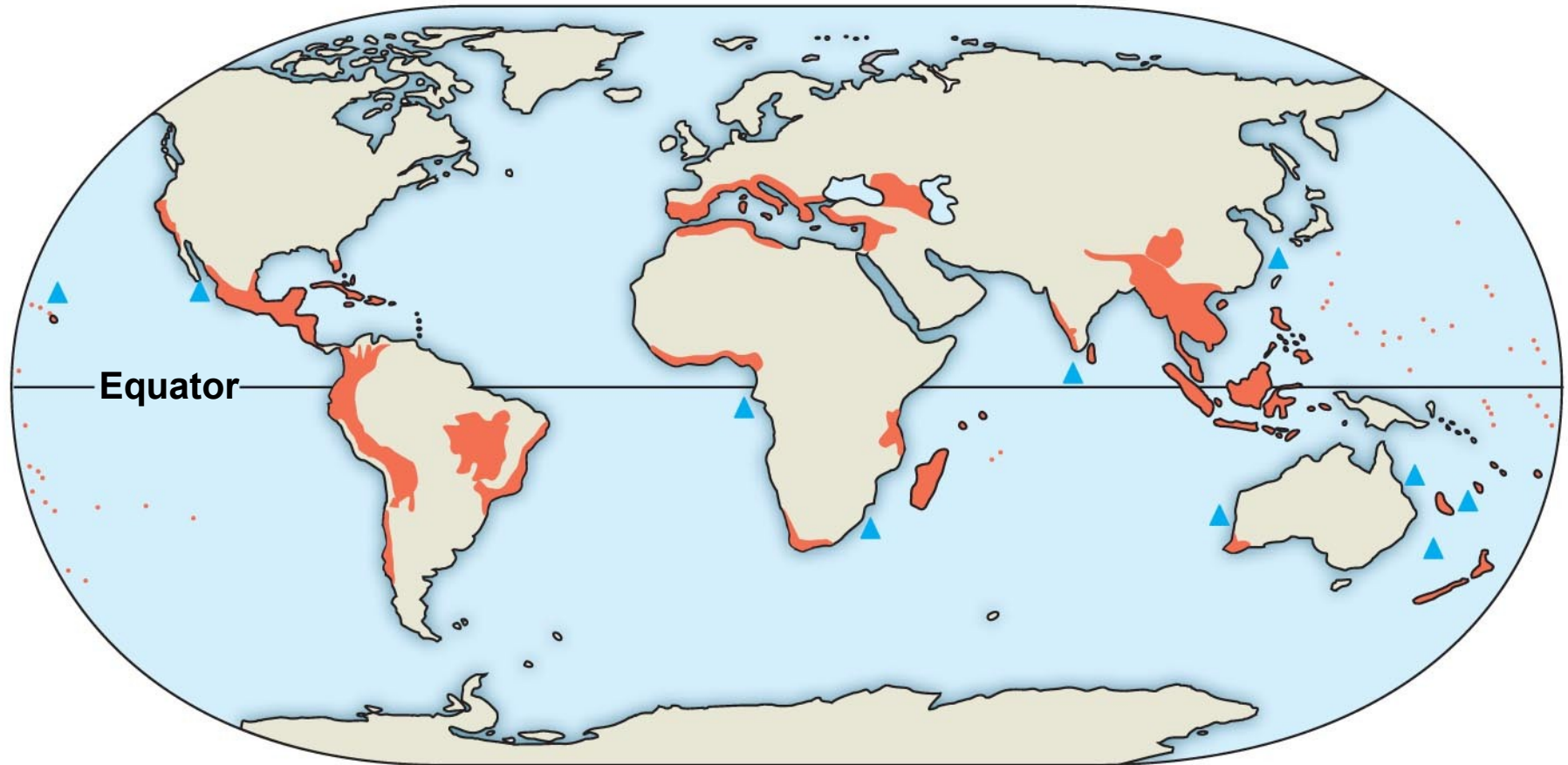
Establishing Protected Areas

- **Conservation biologists** apply understanding of ecological dynamics in establishing protected areas to slow the loss of biodiversity. Much of their focus has been on hot spots of biological diversity.
- A **biodiversity hot spot** is a relatively **small area** with a **great concentration of endemic species** and **many endangered and threatened species**.
- Biodiversity hot spots are good choices for nature reserves, but identifying them is not always easy.

Earth's terrestrial and marine biodiversity hot spots

■ Terrestrial biodiversity hot spots

▲ Marine biodiversity hot spots



Philosophy of Nature Reserves

- ***Nature reserves are biodiversity islands in a sea of habitat degraded by human activity.***
- Nature reserves must consider disturbances as a functional component of all ecosystems.

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- An important question is whether to create fewer large reserves or more numerous small reserves.
 - One argument for extensive reserves is that large, far-ranging animals with low-density populations require extensive habitats.
 - Smaller reserves may be more realistic, and may slow the spread of disease throughout a population.

Zoned Reserves

- The **zoned reserve** model recognizes that conservation often involves working in landscapes that are largely human dominated.
- A zoned reserve includes relatively undisturbed areas and the modified areas that surround them and that serve as buffer zones.
- Zoned reserves are often established as “**conservation areas**.”
- Costa Rica has become a world leader in establishing zoned reserves.

Zoned reserves in Costa Rica



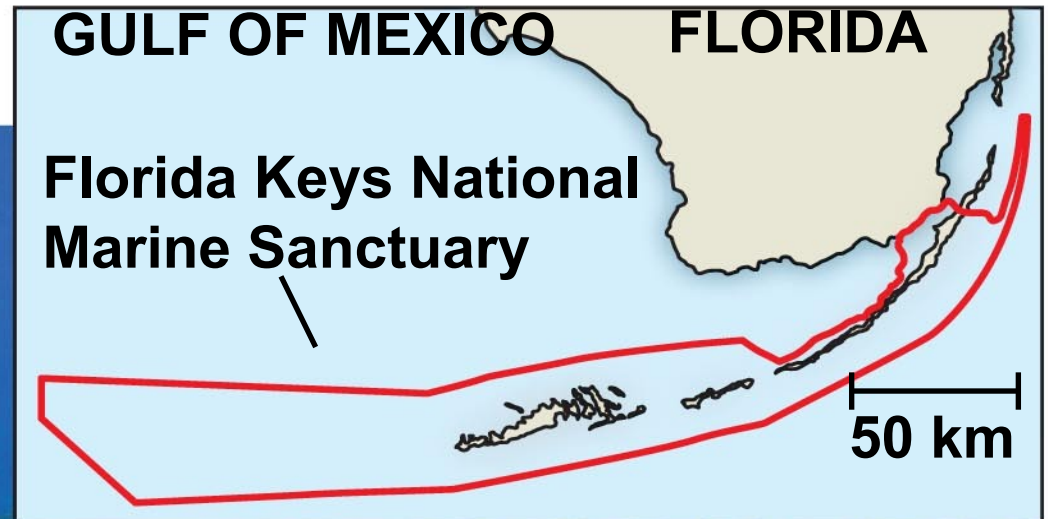
(a) Zoned reserves in Costa Rica



(b) Schoolchildren in one of Costa Rica's reserves

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- Some zoned reserves in the Fiji islands are closed to fishing, which actually improves fishing success in nearby areas.
 - The United States has adopted a similar zoned reserve system with the Florida Keys National Marine Sanctuary.

A diver measuring coral in the Florida Keys National Marine Sanctuary



Restoration ecology attempts to restore degraded ecosystems to a more natural state

- Given enough time, biological communities can recover from many types of disturbances.
- ***Restoration ecology seeks to initiate or speed up the recovery of degraded ecosystems.***
- A basic assumption of restoration ecology is that most environmental damage is reversible.
- Two key strategies are ***bioremediation*** and ***augmentation of ecosystem processes.***

A gravel and clay mine site in New Jersey before and after restoration



(a) In 1991, before restoration



(b) In 2000, near the completion of restoration

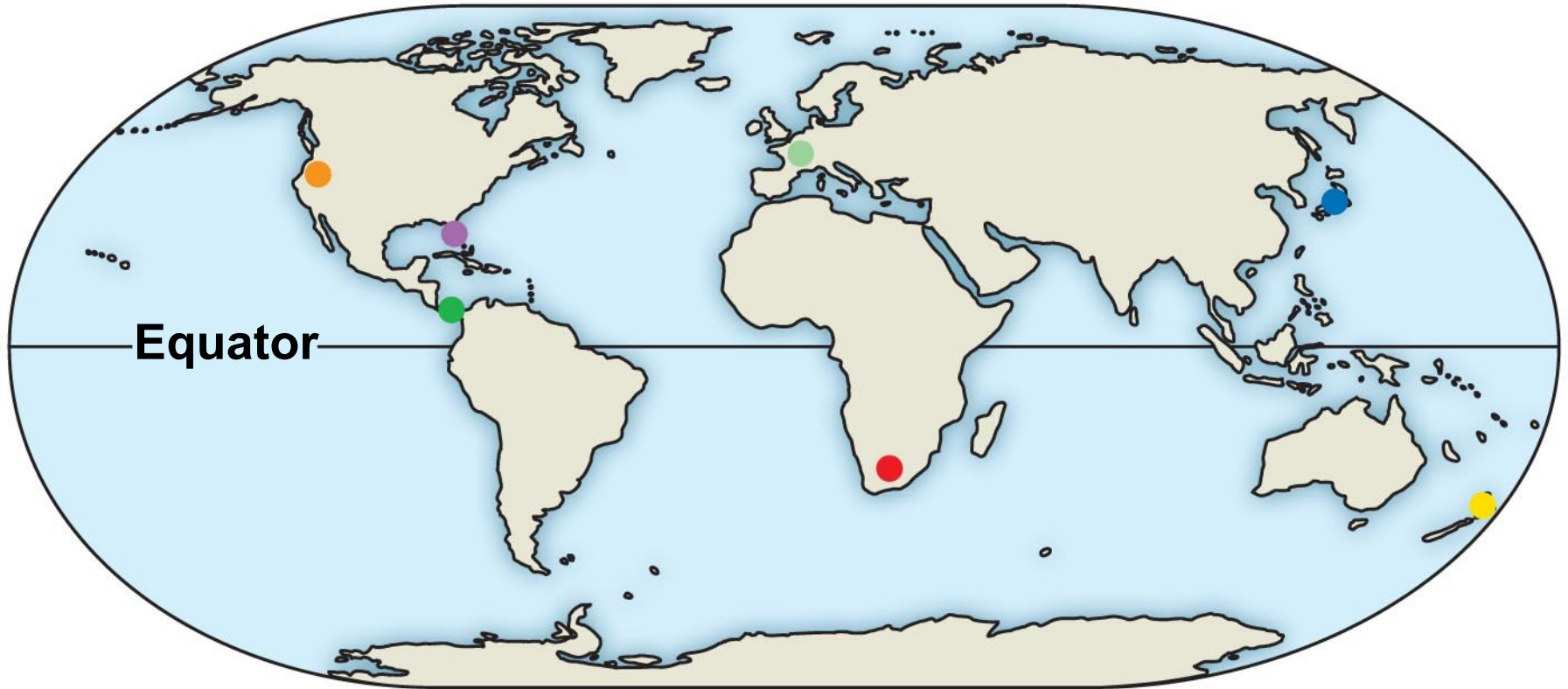
Bioremediation

- **Bioremediation** is the use of living organisms to detoxify ecosystems.
- The organisms most often used are prokaryotes, fungi, or plants.
- These organisms can take up, and sometimes metabolize, toxic molecules.

Biological Augmentation

- **Biological augmentation** uses **organisms** to **add essential materials** to a degraded ecosystem.
- For example, **nitrogen-fixing plants** can increase the available nitrogen in soil.
- The newness and complexity of restoration ecology require that ecologists consider alternative solutions and adjust approaches based on experience.

Restoration ecology worldwide



Restoration ecology



● **Truckee River, Nevada**

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● **Kissimmee River, Florida**

Restoration Ecology



● Rhine River, Europe

Sustainable development seeks to improve the human condition while conserving biodiversity

- The concept of sustainability helps ecologists establish long-term conservation priorities.
- **Sustainable development** is development that meets the needs of people today without limiting the ability of future generations to meet their needs.
- The goal of the Sustainable Biosphere Initiative is to define and acquire basic ecological information for responsible development, management, and conservation of Earth's resources.

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- **Sustainable development** requires connections between life sciences, social sciences, economics, and humanities.
 - Case Study: Costa Rica's conservation of tropical biodiversity involves partnerships between the government, other organizations, and private citizens.
 - Human living conditions (infant mortality, life expectancy, literacy rate) in Costa Rica have improved along with ecological conservation.

The Future of the Biosphere

- Our lives differ greatly from early humans who hunted and gathered and painted on cave walls.
- Our behavior reflects remnants of *our ancestral attachment to nature and the diversity of life*—the concept of *biophilia*.
- Our sense of *connection to nature* may motivate realignment of our environmental priorities.

(a) Detail of animals in a 36,000-year-old cave painting, Lascaux, France



Biophilia: past and present

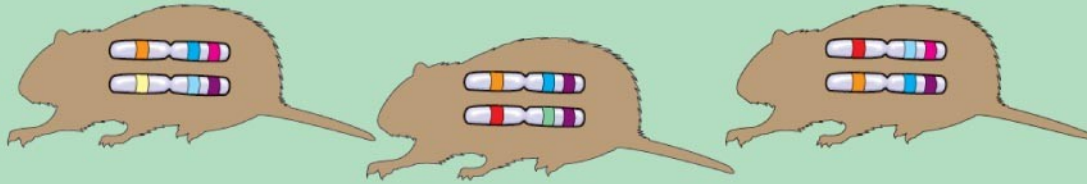


(b) A 30,000-year-old ivory carving of a water bird, found in Germany

(c) Biologist Carlos Rivera Gonzales examining a tiny tree frog in Peru



Review



Genetic diversity: source of variations that enable populations to adapt to environmental changes



Species diversity: important in maintaining structure of communities and food webs



Ecosystem diversity: Provide life-sustaining services such as nutrient cycling and waste decomposition

You should now be able to:

1. Distinguish between conservation biology and restoration biology.
2. List the three major threats to biodiversity and give an example of each.
3. Define and compare the small-population approach and the declining-population approach.
4. Distinguish between the total population size and the effective population size.

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5. Describe the conflicting demands that may accompany species conservation.
 6. Define biodiversity hot spots and explain why they are important.
 7. Define zoned reserves and explain why they are important.
 8. Explain the importance of bioremediation and biological augmentation of ecosystem processes in restoration efforts.

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9. Describe the concept of sustainable development.
 10. Explain the goals of the Sustainable Biosphere Initiative.